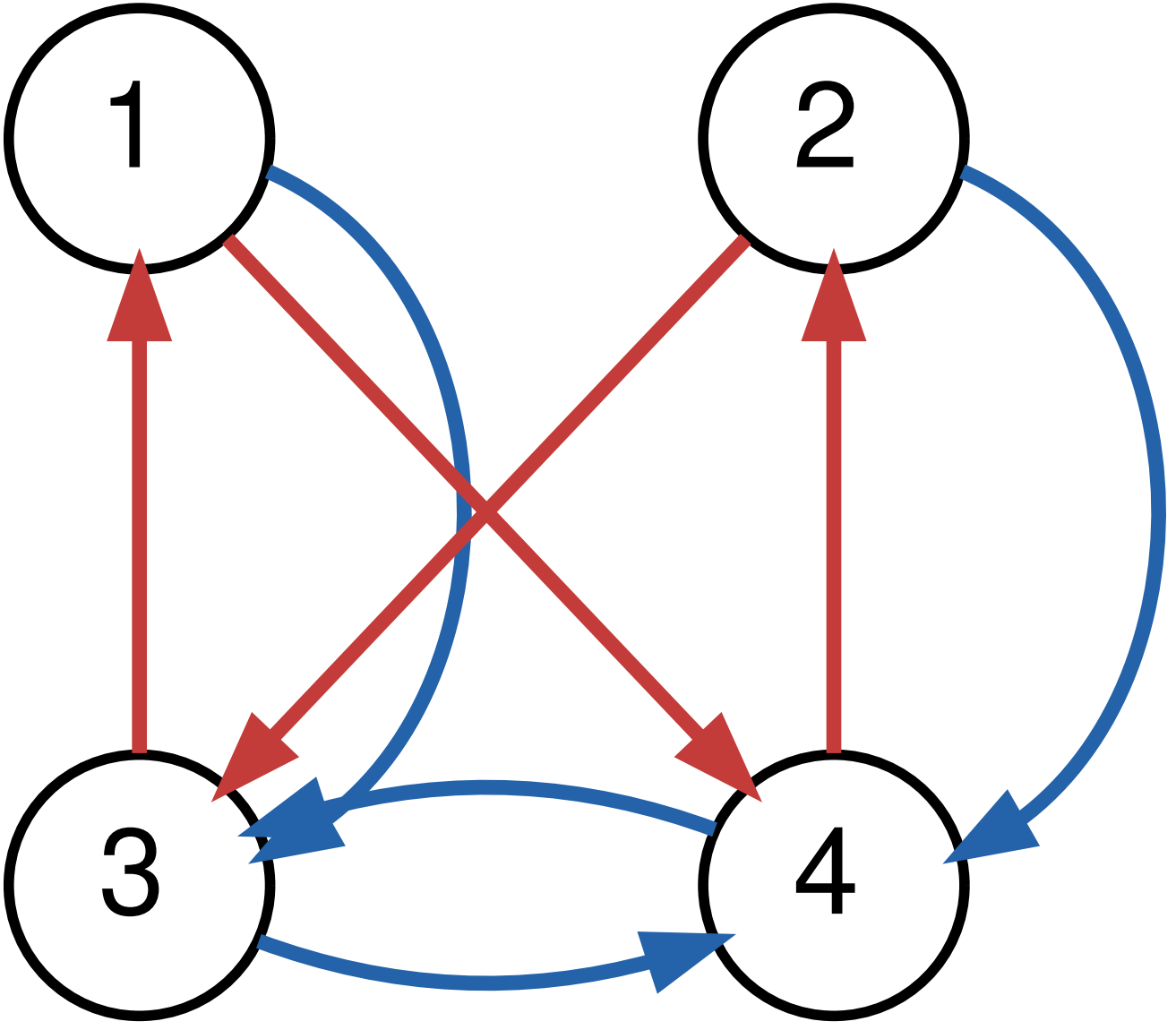(a) Original coloring
R=[4312], B=[3443]